Aleksandr Gyumushyan

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Experience

Software Developer Sep. 2025 – Present
LDA Technologies *Mississauga, ON*

- Replaced a legacy API by engineering a highly concurrent, RPC-like library in Python and Go to communicate with server modules over a serial connection via Protobuf messages. Implemented code generation to streamline new API creation for future products, accelerating their integration into existing tools.
- Ensured data privacy and legal compliance by deploying a private LLM inference stack for sensitive data, benchmarking open-source models on CPU and Multi-GPU systems via vLLM, Llama.cpp, and OpenWebUI with Prometheus and Grafana for monitoring.
- Consolidated two legacy C++ firmware and FPGA image flashing tools into a single Go utility, while to the new API backend, thereby lowering the surface area for bugs and future maintenance.
- Developed 25+ automated hardware and software tests in Python, within a Jenkins Agent framework, covering hardware configuration validation, load testing, service health checks, and system provisioning.

Backend Developer Intern May 2022 – Aug. 2023
Geotab *Oakville, ON*

- Slashed internal Auth API requests by 50%+ across data ingestion gateway workers by engineering a Redis caching layer to eliminate redundant token requests and implement proper refresh mechanics.
- Accelerated vehicle signal build velocity by 99% by co-developing a Python CLI tool to parse vehicle signal profiles into Protobuf structures and firmware-compatible C macros; scaled the tool across the firmware department as a core utility for all new vehicle support.
- Owned the architecture for the tool's enterprise access-control and logging layer. Coordinated with stakeholders and legal on logging compliance, integrating Google OAuth and a BigQuery table for security and usage tracking.

Projects

Ultra-fast 1 Billion Row Challenge | Repo | Writeup | *Zig, Samply* Jan. 2026

- Engineered a high-performance data ingestion engine in Zig to parse a 15GB dataset (1 billion records), achieving a single-core processing time of 9.3 seconds
- Reduced execution time by over 99.9% from a naive solution with a variety of strategies such as: bypassing standard I/O with zero-copy memory mapping, a custom hash table with linear probing, minimizing branching and for loops as well as meticulous benchmarking and testing to verify change correctness and performance
- Explored operating system API functionality such as Memory mapping, demand paging and buffered reading to optimize disk read speeds which was the biggest bottleneck for most of the project

Education

Toronto Metropolitan University Toronto, ON
Bachelor of Science in Computer Science - 3.92 GPA *Sep. 2020 – May 2025*

Technical Skills

Languages: Go, Python, Java, C#, TypeScript, Zig
Databases and Infrastructure: Redis, RabbitMQ, Protobuf, gRPC
Developer Tools and Testing: Git, Jenkins, Linux, Docker